

System Architecture, Module Progress & Strategic Engineering Roadmap

Evaluated By: UI Finish-Gate Reviewer & Senior Engineering Architect | Date: September 2026 | Entity: PITB / GoAJK

Executive Summary

The PMCC CMS AJK platform has achieved an advanced, high-integrity implementation status. The platform adheres rigorously to statutory administrative governance (48-hour acknowledgment, 15-day resolution), high-efficiency golden-ratio layout architectures, and strict database-level security policies. All core workflows—from bilingual citizen complaint registration and dual-factor tracking to authenticated department triage, human-verified AI deduplication, and cascade-protected complaint resolution—have been built with production-grade craftsmanship.

Comprehensive Module Progress Audit

Module & Scope	Primary Domain / Focus	Status	Architectural & UX Implementation Highlights
Module 1: Core Schema	Foundational Database, Relational Constraints, Spatial Entities	100% Complete	- Relational integrity with restrict-on-delete rules - Spatial taxonomy (Districts & Tehsils with native Urdu names) - Audit trail logging via complaint_status_history - pgvector embeddings foundation for deduplication
Module 2: Auth & Access (PRD 1)	Authentication, Inactive Lockout, Role Protection, 2-Step OTP	95% Complete	- Inactive account lockout (is_active=false) - No public self-registration routes - Step 2 OTP screen with 6-box auto-advance, backspace reversal, arrow traversal, paste splitting & live countdown timer - Department-scoping policy against URL tampering
Module 3: Citizen Portal (PRD 2)	Bilingual Public Intake, Camera/Video Capture, Dual-Factor Tracking	95% Complete	- Native RTL/LTR Urdu & English dynamic toggle - Multi-step guided wizard with progress indicators - In-form camera & video capture (60s limit) 24h CNIC spam rate-limiting & deduplication - Dual-factor public tracking (CNIC + Complaint #) - Printable official acknowledgment docket
Module 4: FP Portal (PRD 3)	Department Queue, Investigation, AI Duplicate Review, Resolution	90% Complete	- Golden-ratio Focal Person Dashboard matching prototype - Full unmasked citizen identity for case handling - AI similarity score badges & human confirmation/dismissal 4-Path Classification (Direct, Club, Forward, Field Visit) - Supervised field officer delegation - Cascade resolution confirmation modal for clubbed cases - Cross-department reassignment request mechanism
Module 5: Admin & Director (PRD 4)	System Oversight, 'Others' Queue, Reference Data CRUD, Director Approval	40% Complete	- Global Admin Dashboard controller and metrics visualization - Department performance breakdown and audit logging - Pending: Director reassignment approval/rejection screen, Admin 'Others' manual routing queue, and Reference Data CRUD

Technical Audit & Finish-Gate Review Findings

1. Design Architecture & Craftsmanship

- **Palette & Token Consistency:** Adheres strictly to the AJK Government PMCC identity: Forest Green (#24402F), Warm Sand (#FAF6EF), Heritage Gold (#E8B14A), and Alert Vermilion (#EE3B23).
- **Anti-Generic UI Enforcement:** Rejects generic SaaS aesthetics. Replaces interchangeable cards with statutory SLA priority indicators, complaint lifecycle timelines, and dense, scannable queues.
- **Human-in-the-Loop Governance:** AI similarity matches provide advisory assistance; no child complaint is clubbed or altered without explicit human confirmation, preserving complete legal accountability.

2. Observations Requiring Polish

- **Focal Person Sub-Screen Unification:** While the primary FP Dashboard is fully aligned with the approved golden-ratio prototype, the sub-screens (Investigate.jsx and ComplaintResolve.jsx) should be migrated to the unified FocalPersonLayout.jsx brand shell for visual continuity.
- **Director Reassignment Interface:** The mechanism for requesting cross-department transfers is active; the receiving Director approval view (with mandatory rejection remarks) needs to be completed under Module 5.
- **Admin 'Others' Triage Queue:** Complaints submitted without a defined department require a dedicated triage screen for Admin reassignment.

Strategic Engineering Roadmap & Next Steps

Phase 1: Sub-Screen Brand Unification (Immediate)

1. Migrate Investigate.jsx and ComplaintResolve.jsx to use FocalPersonLayout.jsx, providing persistent navigation tabs, quick stats, and consistent header treatments.
2. Refine the AI duplicate comparison modal with side-by-side text diffing between incoming complaints and potential parent matches.

Phase 2: Completion of Module 5 (Admin & Director Suite)

1. **Director Reassignment Approval Screen:** Dedicated workspace for Directors to approve transfers (updating department_id) or reject them with mandatory review_notes.
2. **Admin 'Others' Triage Interface:** Real-time queue for routing uncategorized complaints with administrative remarks.
3. **System Reference Data Management:** CRUD interfaces for Departments, Sub-departments, Categories, and Forward Destinations.
4. **Staff Account Provisioning:** Secure Admin panel to provision and deactivate Focal Person and Director credentials with one-time temporary passwords.

Phase 3: Production Hardening & Operational Automations

1. **Asynchronous AI Embedding Worker:** Offload embedding generation and similarity indexing to background Laravel queue workers (queue:work).
2. **Statutory SLA Escalation Cron:** Automated scheduled jobs to calculate aging and automatically flag breaches of the 48-hour acknowledgment or 15-day statutory resolution targets.